68320 Nanofabrication and Nanocharacterization Techniques

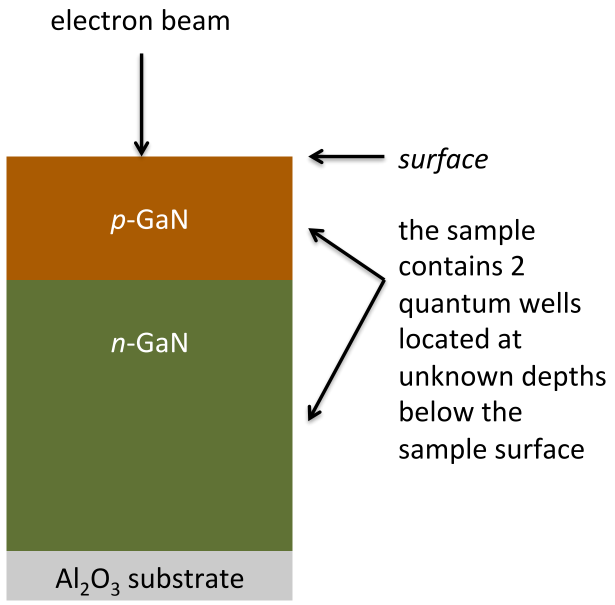
Assignment – Prac Portfolio

Resources for completing this assignment:

- Notes/slides from lectures & tutes.
- A **data file** that contains CL spectra that were collected as a function of electron beam energy: "*CL_data.csv*" (Canvas > Modules > Practicals – Portfolio 1). The spectra are corrected for system response.
- Reading material: The paper "*Depth profiling of GaN by cathodoluminescence microanalysis*" (Canvas > Modules > Practicals > CL Depth profiling.pdf).
- Extra reading material:
  CL review paper (Canvas > Modules > Practicals > CL review paper.pdf).

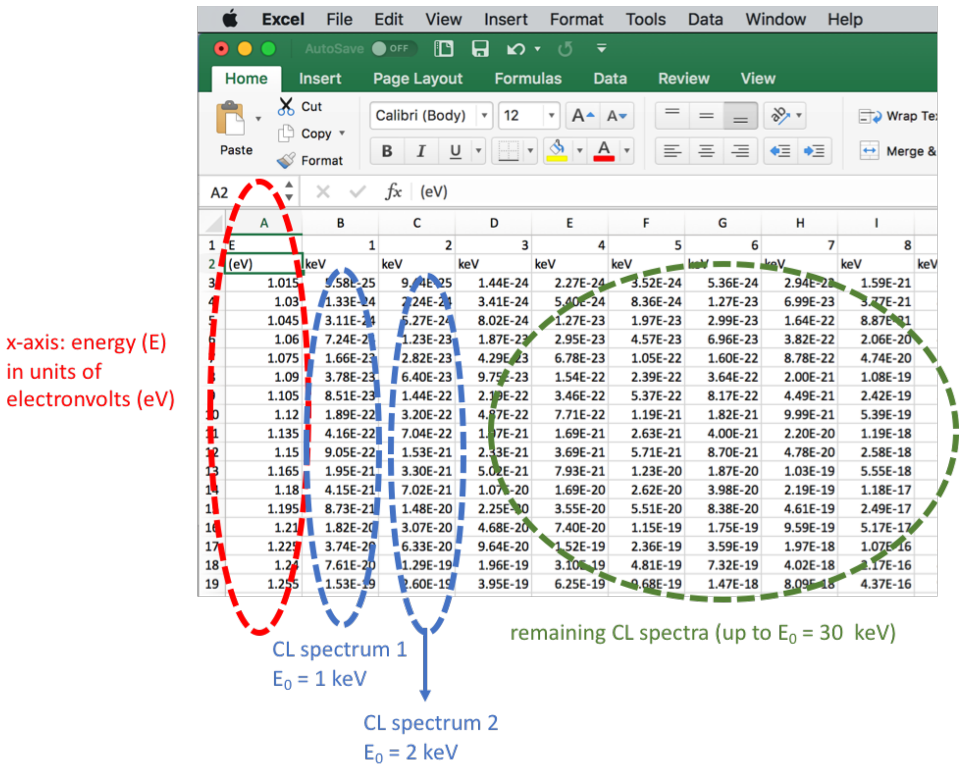
BACKGROUND

A multi-layer structure made from the semiconductor gallium nitride (GaN) was analysed by cathodoluminescence (CL) spectroscopy. A cross-sectional schematic of the sample is shown below:



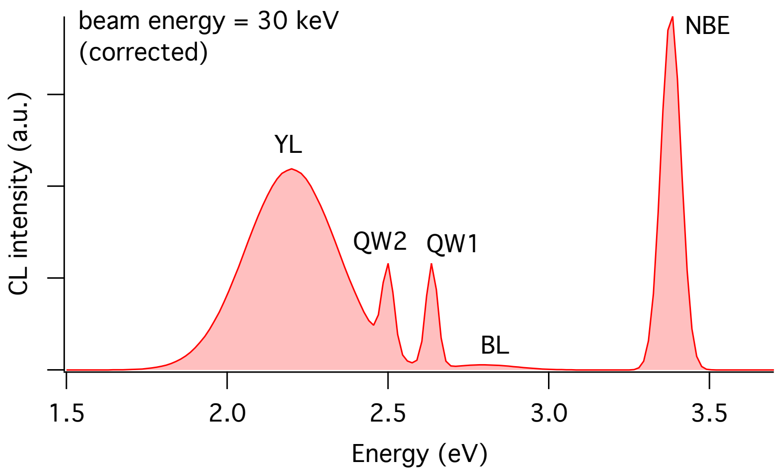
In the above, "*p*" and "*n*" denote p-type doping and n-type doping, respectively. A quantum well is a very thin, 2-dimensional (2D) semiconductor with a bandgap that is smaller than that of GaN.

To perform depth-resolved CL analysis of the sample, CL spectra were acquired as a function of beam energy. The CL signal was collected using a parabolic mirror located above the sample surface. The first spectrum was taken using an electron beam energy (E_o) of 1 keV and a current (I_0) of 20 nA. In all subsequent spectra, the beam power (E_0I_0) was constant. All CL experimental data are saved in the file "**CL_data.csv**" which can be opened using software packages such as Microsoft Excel:



	A	B	C	D	E	F	G	H	I
1	E	1	2	3	4	5	6	7	8
2	(eV)	keV	keV	keV	keV	keV	keV	keV	keV
3	1.015	5.58E-25	9.44E-25	1.44E-24	2.27E-24	3.52E-24	5.36E-24	2.94E-23	1.59E-21
4	1.03	1.33E-24	2.24E-24	3.41E-24	5.40E-24	8.36E-24	1.27E-23	6.99E-23	3.77E-21
5	1.045	3.11E-24	5.27E-24	8.02E-24	1.27E-23	1.97E-23	2.99E-23	1.64E-22	8.87E-21
6	1.06	7.24E-24	1.23E-23	1.87E-23	2.95E-23	4.57E-23	6.96E-23	3.82E-22	2.06E-20
7	1.075	1.66E-23	2.82E-23	4.29E-23	6.78E-23	1.05E-22	1.60E-22	8.78E-22	4.74E-20
8	1.09	3.78E-23	6.40E-23	9.75E-23	1.54E-22	2.39E-22	3.64E-22	2.00E-21	1.08E-19
9	1.105	8.51E-23	1.44E-22	2.19E-22	3.46E-22	5.37E-22	8.17E-22	4.49E-21	2.42E-19
10	1.12	1.89E-22	3.20E-22	4.87E-22	7.71E-22	1.19E-21	1.82E-21	9.99E-21	5.39E-19
11	1.135	4.16E-22	7.04E-22	1.07E-21	1.69E-21	2.63E-21	4.00E-21	2.20E-20	1.19E-18
12	1.15	9.05E-22	1.53E-21	2.33E-21	3.69E-21	5.71E-21	8.70E-21	4.78E-20	2.58E-18
13	1.165	1.95E-21	3.30E-21	5.02E-21	7.93E-21	1.23E-20	1.87E-20	1.03E-19	5.55E-18
14	1.18	4.15E-21	7.02E-21	1.07E-20	1.69E-20	2.62E-20	3.98E-20	2.19E-19	1.18E-17
15	1.195	8.73E-21	1.48E-20	2.25E-20	3.55E-20	5.51E-20	8.38E-20	4.61E-19	2.49E-17
16	1.21	1.82E-20	3.07E-20	4.68E-20	7.40E-20	1.15E-19	1.75E-19	9.59E-19	5.17E-17
17	1.225	3.74E-20	6.33E-20	9.64E-20	1.52E-19	2.36E-19	3.59E-19	1.97E-18	1.07E-16
18	1.24	7.61E-20	1.29E-19	1.96E-19	3.10E-19	4.81E-19	7.32E-19	4.02E-18	2.17E-16
19	1.255	1.53E-19	2.60E-19	3.95E-19	6.25E-19	9.68E-19	1.47E-18	8.09E-18	4.37E-16

A plot of the final spectrum in the file (beam energy, E_0 = 30 keV) is shown below:



The spectrum contains 5 CL emission peaks:

- NBE “near-bandgap emission”
- BL “blue luminescence”
- QW1 “quantum well 1”
- QW2 “quantum well 2”
- YL “yellow luminescence”

ASSIGNEMNT

Write a report in the form of a scientific paper that describes the CL experiment that was performed to collect this data, and what you learned from your analysis of the data. The report should have these sections:

- Title
- Abstract

  A succinct summary of your report suitable for a scientist who is not an expert in this field of research.

- Introduction
  Introduction to the field that lays out the context for your report. This should include a brief literature review with references to key papers in the field.
- Experimental methodology
  A description of the microscope used for this work (this can include a schematic and a photograph of an SEM), and the procedure used to collect the data in the report.
- Results and Discussion
  Results, and your analysis of the results and a discussion of what you have learnt.
- Conclusion
  Similar to an abstract, but it can include technical details and can assume domain expertise.

Include the following in your report:

1. Plot all 30 CL spectra.

   NOTE: Make sure all your figures and figure captions are clear and easy to understand. Label all features discussed in the text, axes, etc. Decide carefully when it is appropriate to combine or break up figures. (e.g., is it best to plot 10 spectra on 10 individual graphs, or plot them all on a single graph and use color, vertical offsets, log scales or other plotting techniques to improve clarity?)

2. Use the spectra to generate depth-profiles of all CL emissions in the spectra. Briefly discuss how this may be done. For example, how can *peak fitting* be used to get the most accurate CL depth profiles? Plot all 5 depth profiles on a single graph.

   HINT: A "depth profile" is a plot of CL intensity as a function of beam energy.

3. Estimate how deep each one of the 2 quantum wells is below the sample surface. Explain how you worked this out.

   HINT: You can use this equation:

   $$R_e(nm) = \frac{27.6AE_o^{5/3}}{Z^{8/9}\rho}$$

4. Inspect your depth profiles and think about the emission labelled "NBE" in the above spectrum. Is the NBE depth-profile physically realistic? Explain your answer.

Marking criteria

- Technical & scientific accuracy – 50%. Be very precise with the terminology that you use in your report. Scrutinize every sentence and ensure the physics is correct.
- Presentation – 50 %. The report must feel professional. The text should be succinct but complete. There is no upper or lower size limit, but redundant bloat will be penalized. If you do have detailed information that you want to include (e.g. code that you have written to do data analysis), consider the use of an appendix, or provide a link to your code in a repository or a cloud. Ensure your figures are of publication quality.

Ask your lecturer if you have any questions about expectations and marking criteria – the ability to query, understand and deliver on expectations is a key skill to success in the workplace.